

Problem

Calculate $|H(\omega)|$ if H_{dB} equals

- (a) 0.1 dB
- (b) -5 dB
- (c) 215 dB

Step-by-step solution

Step 1 of 3

(a)

Write the relation between H_{dB} and $|H(\omega)|$.

$$H_{dB} = 20 \log_{10} |H(\omega)|$$

Substitute 0.1 for H_{dB} .

$$0.1 = 20 \log_{10} |H(\omega)|$$

$$\begin{aligned} \log_{10} |H(\omega)| &= \frac{0.1}{20} \\ &= 0.005 \end{aligned}$$

$$\begin{aligned} |H(\omega)| &= 10^{0.005} \\ &= 1.01 \end{aligned}$$

Therefore, the value of $|H(\omega)|$ is $\boxed{1.01}$.

Step 2 of 3

(b)

Write the relation between H_{dB} and $|H(\omega)|$.

$$H_{dB} = 20 \log_{10} |H(\omega)|$$

Substitute -5 for H_{dB} .

$$-5 = 20 \log_{10} |H(\omega)|$$

$$\begin{aligned} \log_{10} |H(\omega)| &= -\frac{5}{20} \\ &= -0.25 \end{aligned}$$

$$\begin{aligned} |H(\omega)| &= 10^{-0.25} \\ &= 0.562 \end{aligned}$$

Therefore, the value of $|H(\omega)|$ is $\boxed{0.562}$.

Step 3 of 3

(c)

Write the relation between H_{dB} and $|H(\omega)|$.

$$H_{dB} = 20 \log_{10} |H(\omega)|$$

Substitute 215 for H_{dB} .

$$215 = 20 \log_{10} |H(\omega)|$$

$$\begin{aligned} \log_{10} |H(\omega)| &= \frac{215}{20} \\ &= 10.75 \end{aligned}$$

$$\begin{aligned} |H(\omega)| &= 10^{10.75} \\ &= 5.623 \times 10^{10} \end{aligned}$$

Therefore, the value of $|H(\omega)|$ is 5.623×10^{10} .